

Hunger related diseases

Global Challenge Group 9



Introduction

Hunger can cause diseases directly or indirectly. Although it is rarely listed as the direct death cause, it can account for great many deaths(1). The diseases are widely reported in history. For example, in Amsterdam during World War 2, a famine led to childhood malnutrition, resulting in adult pathologies including an increased propensity to heart disease and diabetes(2).

Today, many people still suffer from hunger. Thus, understanding hunger related diseases can be of great help. In this poster, we will explain the progress, major symptoms, consequences and general treatment of hunger-related diseases and try to find out some helpful solutions.

Starvation

Starvation is one of the extreme cases of undernourishment. In a hunger, people usually starve. Starvation can induce a series of biological changes that will eventually exhaust your body by consuming glucose, proteins and other nutrients.

Glycogenolysis

Induced by insulin decline and low blood sugar, our body begins to release and consume the sugar stored in the liver and muscle named glycogen. But this storage can only last 24-48 hrs. (4)

Gluconeogenesis

Soon, our body begins to reuse amino acids, glycerol and lactate to synthesize glucose. This process will happen in the first few days of starvation and then declines. (4)

Ketogenesis

Finally, low insulin causes ketogenesis. Tissues begin to use ketones, which can cause poisoning, as the energy source. Liver will use fat to produce ketones (4).

Death

If the starvation continues, how long you can keep alive is up to your fat storage and the micronutrient intake. After 60-70 days, death may well happen, yet micronutrient intake can extend the time. (4)

Catabolysis is happening and damaging the neural system during the process! Protein insufficiency will cause **Kwashiorkor** and **Marasmus**.

Micronutrients

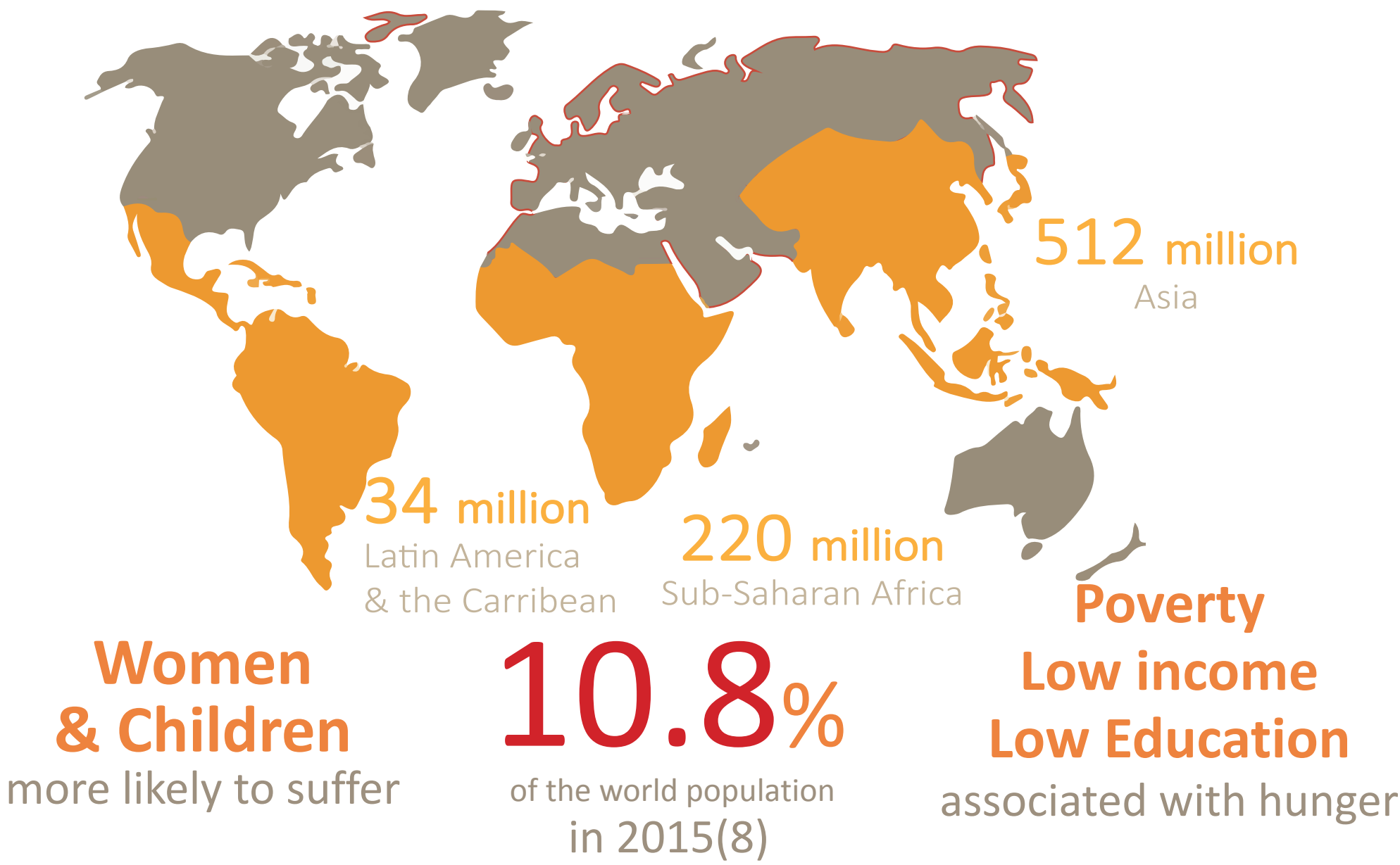
Undernourishment, which means the patients do not acquire enough nutrition from the diets, can cause a lot of health problems(5). Starvation is one of the extreme case of undernourishment, while other cases of undernourishment are also regarded as **hidden hunger**(3). Micronutrients are important to metabolism and could **relieve the harm** during starvation. It was found that a starving obese man with enough micronutrients lived as long as a whole year. (4) Insufficient micronutrients can lead to various diseases as listed below:

Mineral Defficiency	Vitamin Defficiency
Iron deficiency	Vitamin B Riboflavin deficiency Pellagra (Niacin deficiency)
Iodine deficiency	...
...	Other vitamins ...

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Who is suffering?



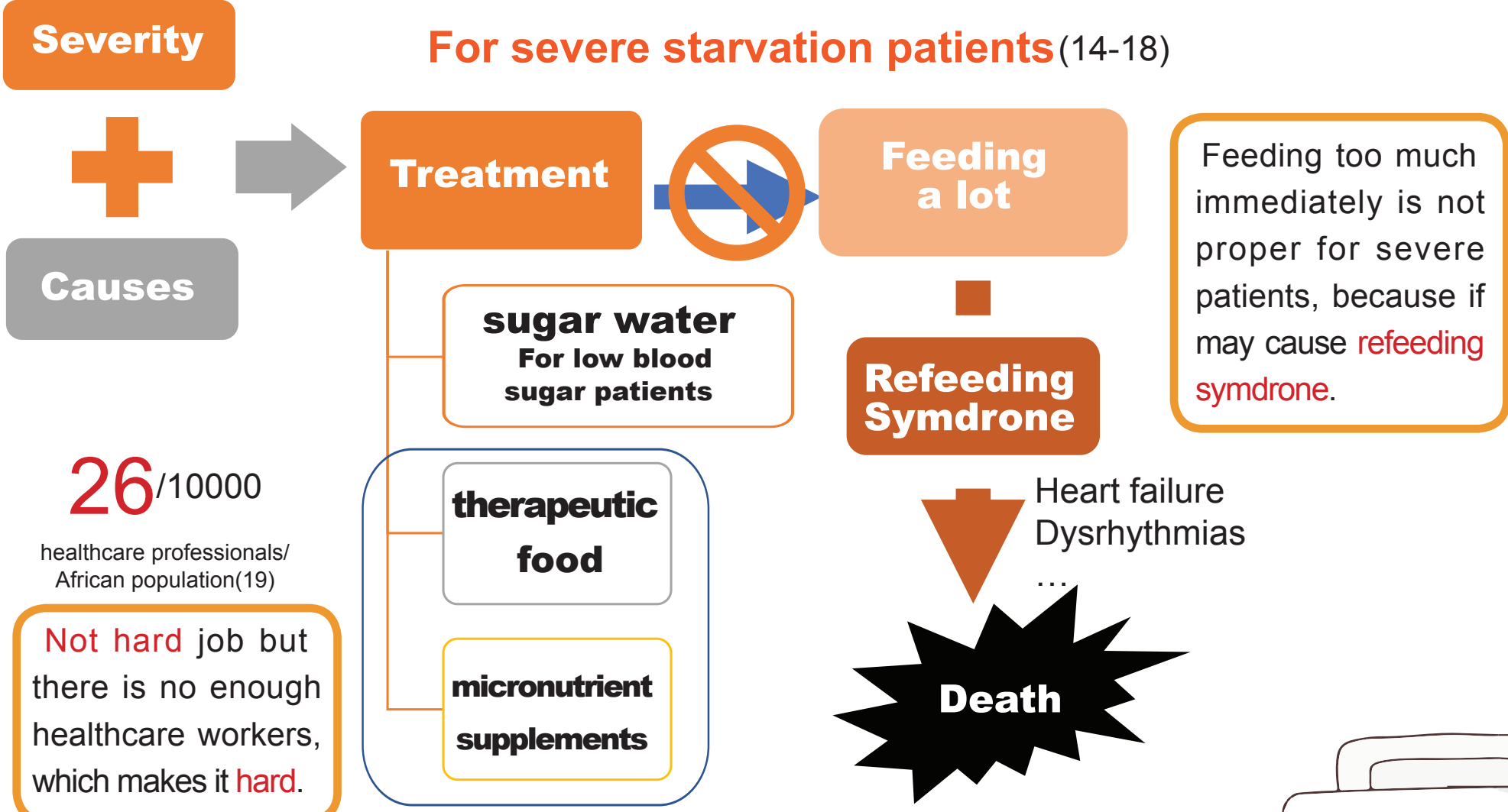
Symptoms

- 1.low blood sugar & low insulin
- 2. stomach atrophy
- 3.loss of weight
- 4.fatigue
- 5.menstruation ceases
- 6.dehydration
- 7.edema and so on...

Consequences

- 1.Memory loss
Severe hunger makes brain cannot get enough glucose. Ketone is then used as energy source (10), leading to poisoning and possibly causing memory decline or even memory loss.
- 2.Gallstone
Eating few meals per day and inadequate intake of necessary nutrients are both risk factors for the formation of gallstone (11).
- 3.Ageing
Glycogen, fats and even proteins may be composed, which is likely to induce dry skin, wrinkling (12) or anemia, causing other ageing related disease. (13).
- and so on...

General Treatment



Possible Solutions

- Long term resolution**
 - Improve agricultural practices
 - Reduce poverty
 - Improve sanitation
 - The empowerment of women
- Short term efforts**
 - Sending professional workers
 - Training more workers for health care in Africa



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